

## **L01 Exploring Real-World Applications of Computer Vision:**

### **Computer Vision in Medical Image Analysis**

Medical image analysis is one of the more practical uses of computer vision because the inputs are already visual, often digitized, and somewhat standardized. The goal in these applications is to pull clinically useful information out of images faster, more consistently, or sometimes more accurately than traditional human review workflows.

#### **Description of the Application**

Medical image analysis has the potential to improve efficiency and consistency in healthcare. It is also already a rapidly developing field. The FDA maintains a list of “Artificial Intelligence-Enabled Medical Devices,” but the agency notes that this list is mainly meant to help innovators understand the current device landscape and to inform the public when AI is being used in medical devices that have been cleared for use in the United States. Inclusion on the list does not necessarily mean that a device has been extensively tested in real-world clinical scenarios or proven to be clinically rigorous in every setting (U.S. Food and Drug Administration).

#### **Technology Behind It**

The common pipeline is to collect labeled medical images, preprocess them, train a model, validate the model, and then deploy it into a clinical workflow. First, images are collected from medical or analytical devices. Before analysis, images are often preprocessed into a more consistent format so the model can process them more reliably. Then, deep learning models are trained to recognize image features connected to tasks such as classification, object detection, segmentation, diagnosis, or measurement. Segmentation means identifying and outlining a specific region of interest, such as an organ, tumor, lesion, vessel, or other abnormal area (Litjens et al.; Singh et al.; Barragán-Montero et al.).

Computer vision does the actual image interpretation work. For example, classification models can decide whether an image is likely normal or abnormal. Object detection models can draw boxes around possible findings. Segmentation models can outline an organ, mass, vessel, or region of concern pixel by pixel. These systems are often paired with other technologies, including hospital imaging archives, electronic health records, regulatory software, and user interfaces that show heat maps or regions of interest.

## **Benefits and Challenges**

One of the biggest benefits is workload reduction. A 2024 meta-analysis of 36 studies found that, for radiologists, AI assistance reduced image reading time by 27.20 percent, and AI pre-screening reduced reading quantity by 61.72 percent in the pooled analysis (Chen et al.). This is significant because humans are capable of getting tired, distracted, and inconsistent. If used carefully, this could improve quality of life for clinicians while also helping standardize results.

The challenge is that performance in a paper or regulatory submission does not automatically equal safe performance in the real world. In the same field of radiology that has seen such benefit, a comprehensive review from 2025 found that among 717 radiology devices with submission documentation, only 33 devices, or 5 percent, had prospective testing; 56, or 8 percent, included human-in-the-loop testing; and 208, or 29 percent, incorporated clinical testing (Sivakumar et al.). These statistics are alarming. If the product is going to be used with a human, it should be tested with a human. If it is going to affect clinical care, it should undergo much more rigorous testing.

## **Reflection**

My own work experience shapes my opinion. I work as a research scientist at a genetics company, and AI or automation is often floated as a way to speed up interpretation. I have done analysis work on hundreds of thousands of samples where calls are made on data in a very manual way. My rough experience is that about 95 percent of data is very standard, about 4 percent is anomalous in a way that could probably be programmed for, and about 1 percent needs real human reasoning. That 1 percent may involve additional context, a weird edge case, or a decision where the pattern alone is not enough. In medically significant work, that is exactly the group I would not want to hand entirely to an automated system that relies on previously observed patterns instead of novel reasoning.

The future of medical image analysis will probably move toward larger multimodal systems that combine images with reports, clinical history, laboratory values, genetics, and carefully crafted patient wellness profiles. This could be genuinely useful, because many hard medical decisions are not image-only decisions. However, I believe that clinically significant decisions that affect people's lives and health should require a human in the loop to sign off on the data and view the images, rather than handwaving an AI-generated decision through.

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### Works Cited

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